

Shoaib Asghar

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EDUCATION

B.S. Computer Science (Graduated) | Information Technology University Punjab | Aug 2022 – Jun 2026

Relevant Coursework: Data Structures & Algorithms, OOP, Database Systems, AI, Machine Learning, Operating Systems

PROFESSIONAL SUMMARY

Backend-focused Python Developer with professional experience building high-performance REST APIs, routing systems, and data pipelines using Django, Django REST Framework, and FastAPI. Skilled in backend architecture, geospatial data processing, API integration, and database engineering with PostgreSQL. Experienced in Agile delivery within production engineering environments.

EXPERIENCE

Python Backend Developer Intern | Neuralige | Feb 2026 – Jun 2026

Tech: Python, Django 5, Django REST Framework, FastAPI, PostgreSQL, OpenRouteService, Geopy, Pandas, LangChain

- Architected and delivered the Fuel Route Optimizer — a production-grade RESTful API using Django REST Framework, integrating OpenRouteService for multi-stop route computation and Geopy for geocoding to optimize travel costs.
- Implemented server-side caching strategies within the routing API to minimize redundant external API calls and improve average response times significantly.
- Built CLARA, an enterprise RAG backend integrating LangChain, ChromaDB vector database, and OpenRouter LLM APIs with a real-time Tavily-powered web search pipeline.
- Contributed SQL-driven data workflows to RankOne enterprise platform, demonstrating strong relational database design skills applicable across Django ORM and SQLAlchemy.

Backend Developer Intern | AppifyTech | Jun 2025 – Aug 2025

Tech: FastAPI, Django, Flask, PostgreSQL, NumPy, Pandas, Scikit-learn, JWT, Git

- Developed and deployed multiple backend services using FastAPI, Django, and Flask, implementing RESTful endpoints with JWT authentication and role-based access control.
- Built ML data processing pipelines using NumPy, Pandas, and Scikit-learn to preprocess and transform datasets for model inference within backend workflows.
- Designed and optimized PostgreSQL schemas, writing complex queries and leveraging indexing to support high-throughput API endpoints.

PROJECTS

Fuel Route Optimizer — *Python, Django 5, DRF, OpenRouteService API, Geopy, Pandas*

- Designed a fuel-cost optimization API that geocodes multi-stop itineraries, queries OpenRouteService for optimal routes, and computes cost savings — handling edge cases and API rate limits robustly.
- Applied Pandas for route data transformation and implemented caching middleware to optimize API performance under repeated query patterns.

CLARA — Enterprise AI RAG Assistant — *LangChain, RAG, LLMs, ChromaDB, Python*

- Built an enterprise-grade Python backend integrating LangChain orchestration, ChromaDB vector store, and OpenRouter LLMs — demonstrating advanced API integration and backend service design.

Smart Car Assistant (FYP) — *Python, OpenCV, YOLOv8, Raspberry Pi*

- Developing a Python-based real-time computer vision pipeline on embedded hardware, showcasing systems-level Python programming and performance optimization.

Facial Expression Recognition Network — *Python, NumPy, Pandas, Deep Learning*

- Engineered a neural network from scratch using NumPy, achieving 85%+ test accuracy — demonstrating deep understanding of data pipelines, matrix operations, and Python numerical computing.

Airbnb Web Architecture Clone — *React.js, Node.js, Express.js, MongoDB, JWT*

- Built full-stack REST APIs with Node.js/Express.js and MongoDB, demonstrating backend API design skills across multiple technology stacks.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C/C++, Bash

Backend Frameworks: Django, Django REST Framework, FastAPI, Flask

Databases: PostgreSQL, MongoDB, ChromaDB, SQL Server

Data & ML: Pandas, NumPy, Scikit-learn, LangChain

APIs & Integration: REST APIs, OpenRouteService, Geopy, Tavily, OpenRouter, JWT

Tools: Git, GitHub, Docker, Postman, Linux/Bash, Jira